



USE CASES FOR EMERGING TECHNOLOGIES TO STRENGTHEN HIGH-CONTAINMENT LABORATORY GOVERNANCE

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Rapid advancements in biotechnology and digital sciences are transforming possibilities for producing biological weapons. The ways in which researchers develop, produce and share biological materials are being redefined by tools such as artificial intelligence (AI) and distributed ledger technology (DLT). This has raised the spectre of biotechnology being misused and malicious actors obtaining biological weapons. Yet, the influence of emerging technologies on how the scientific community conducts and governs biological research also offers promising opportunities to strengthen oversight and transparency across the international regime for the prohibition of biological weapons, which is predominantly centred around the 1972 Biological Weapons Convention (BWC).¹

This research policy paper examines how emerging technologies intersect with biological research in high-containment laboratories (HCLs)—facilities designed to safely handle pathogens that cause infectious disease.² These laboratories represent indispensable elements of both health security and the governance of dual-use science (i.e. science with both civilian and military applications). HCLs enable research within secured environments to support development of vaccines and therapeutics and provide advanced diagnostic capacity for both routine disease surveillance and outbreak response. They thus contribute to national implementation of the BWC by providing the infrastructure and oversight needed to ensure that work with dangerous pathogens remains safe, secure and directed towards peaceful purposes.³

Technological innovation has expanded both the capabilities and the risks associated with research in HCLs. This paper explores how these technological advances interact with the regime for the prohibition of biological weapons. It first examines the expanding role of HCLs and the oversight challenges that they pose. It then outlines three use cases for emerging

SUMMARY

● High-containment laboratories (HCLs) are expanding worldwide as countries invest in infectious disease research and health security. These facilities are essential for developing vaccines, therapeutics and diagnostics, but their rapid growth has outpaced transparency and consistent oversight. Confidence in compliance with the 1972 Biological Weapons Convention (BWC) is weakened by these variations in regulatory capacity and reporting.

Emerging technologies can help address these challenges. Artificial intelligence (AI) and distributed ledger technology (DLT) can strengthen national biosafety systems through enhanced data integrity, traceability and transparency in HCLs. Three use cases illustrate prospective applications: AI-enabled DNA screening to manage synthesis risks; safeguarding laboratory data and records with AI and DLT to improve accountability; and DLT-backed provenance tracking to monitor pathogen materials.

Embedded within national oversight frameworks, these tools could enhance transparency and collaboration through more standardized reporting. In this way they could reinforce biological weapon prohibition norms and build confidence in peaceful intent.

¹ Convention on the Prohibition of the Development, Production and Stockpiling of Bacteriological (Biological) and Toxin Weapons and on their Destruction (Biological Weapons Convention, BWC), opened for signature 10 Apr. 1972, entered into force 26 Mar. 1975, British Foreign and Commonwealth Office, Treaty Series no. 11 (1976).

² On use of AI and DLT in the export control of dual-use biotechnology see Brockmann, K. and Héau, L., 'Use cases for emerging technologies to strengthen export controls on biological items', SIPRI Research Policy Paper, Dec. 2025.

³ Biological Weapons Convention (note 1), Article II.



technology that could improve transparency and accountability in high-containment research. The paper closes by considering how sustained collaboration between the scientific and security communities could strengthen confidence in compliance with the BWC.

I. High-containment laboratories

National biosafety and biosecurity frameworks combine legal, regulatory and institutional measures that ensure safe and secure management of biological materials. They include standards for laboratory practices, containment and oversight. Approaches to these frameworks necessarily reflect local public health priorities, resource constraints, and linguistic or institutional differences. There is no single global model; what matters is functional equivalence and sustained engagement.

Governance of laboratories is a core element of biosecurity as these facilities increasingly serve as access points to dual-use capabilities. High-containment laboratories are classified as biosafety level 3 (BSL-3) or biosafety level 4 (BSL-4) based on the level of containment required to safely handle infectious agents. BSL-3 facilities handle pathogens that can cause serious disease (e.g. West Nile virus or the bacteria responsible for tuberculosis) using enhanced respiratory protection and restricted access. BSL-4 laboratories work with especially dangerous agents capable of causing life-threatening disease (e.g. the Ebola or Marburg viruses) under maximum containment conditions.

The global expansion of HCLs has outpaced the establishment of consistent oversight, which has created uncertainty about how safety and security standards are applied. Three persistent gaps continue to undermine international confidence in safe and peaceful HCL operation.

1. *Transparency mechanisms are weak*, which limits understanding of biosafety and biosecurity practices across borders.
2. *There is no common regulatory compliance model*, which makes it difficult to establish international standards for safety and security.
3. *Oversight capacity is uneven*, with many states lacking sustained resources for inspection, training and enforcement.

Transparency

Visibility into HCL operations has not kept pace with their rapid expansion, which has reduced confidence in peaceful intent. Based on 2024 data, a global mapping study identified 110 BSL-4 laboratories in 34 countries—up from 69 in 2023 and only 13 in 2000—and an estimated 3500 BSL-3 laboratories worldwide.⁴ While this growth in HCLs was primarily in regions with advanced research infrastructure, publicly accessible information on their

⁴ Muluneh, A. G. et al., 'Mapping biosafety level 3 (BSL-3) and BSL-4 laboratories for public health threats reduction', *Journal of Public Health (Berlin)*, published online 20 May 2025; Global BioLabs Initiative, *Global BioLabs Report 2023* (King's College London: London, Feb. 2023); and Millett, P., 'Safeguard the world's worst pathogens', *Science*, 19 Oct. 2023.



locations and activities remains limited. The study found that only 27 per cent of BSL-3 laboratories had any publicly accessible data on their location or the pathogens that they study.⁵ This does not suggest that precise coordinates or pathogen inventories must be disclosed; rather, it reflects the absence of consistent reporting systems that enable governments and international partners to understand where high-risk research capacity exists.

National reporting under the BWC's confidence-building measures (CBMs) is the primary mechanism for states parties to share information on HCLs, including their names, locations and general activities. However, participation in CBM submissions is uneven—around 40 per cent of states parties do not submit reports in any given year—and the level of detail varies considerably among those that do participate.⁶ Some states parties choose not to make their submissions public, further limiting transparency. As a result, information on newly constructed or operational facilities is often incomplete or delayed. This limits opportunities for early dialogue, transparency and the development of shared standards among states parties and has the potential to fuel mistrust. Uneven reporting also underscores the broader challenge of sustaining politically binding but unenforced transparency measures within the BWC.

Regulation and governance

How HCLs are governed has direct implications for international trust in compliance with the BWC. Unlike other arms control treaties, the BWC lacks a verification mechanism to confirm that its states parties are complying with treaty commitments. Oversight therefore depends largely on national systems. However, these systems often differ widely in scope and rigour. Coordination and accountability among states remain limited, particularly in how HCL operations are regulated and monitored. According to the global mapping study, 131 of the 143 countries (91 per cent) with BSL-3 facilities reviewed in 2024 had no national guidance on oversight of dual-use research or on responsible science practices.⁷ While many BWC states parties maintain internal mechanisms or legal measures that are not visible through open-source review, this figure nevertheless suggests that national oversight of these HCLs remains uneven.

Regulatory standards for high-containment facilities also vary significantly. Some states operate formal licensing and inspection regimes, while others rely on informal or outdated regulatory arrangements.⁸ Compliance responsibilities are often split across multiple ministries or agencies—health, science, agriculture, security or defence—without a single coordinating authority. While regulatory diversity is inevitable given differences in language, institutional culture and health priorities, it still complicates mutual understanding and baseline transparency.

⁵ Muluneh et al. (note 4).

⁶ UN Office for Disarmament Affairs, 'Biological weapons: Confidence building measures', [n.d.]; and UN Office for Disarmament Affairs, BWC Implementation Support Unit, 'CBM statistics', BWC Confidence Building Measures.

⁷ Muluneh et al. (note 4).

⁸ Millett (note 4).

Persistent gaps between national oversight of biosafety and biosecurity practices and international mechanisms intended to promote transparency and accountability hinder progress. Working papers submitted to the BWC Working Group on the Strengthening of the Convention and a resolution adopted in 2024 by the World Health Assembly acknowledge these limitations, but practical mechanisms for translating global commitments into coordinated national action remain limited.⁹

These coordination challenges are most visible in states operating high-containment or advanced research facilities, but the underlying principles of biosafety, biosecurity and responsible oversight apply to all research environments. Few low- or middle-income countries currently host BSL-3 or BSL-4 laboratories, but numbers may rise as global capacity expands. Many already operate smaller diagnostic and research facilities that handle potentially hazardous biological materials. Strengthening traceability, documentation and personnel accountability in such facilities can reinforce national implementation of the BWC and support broader public health resilience. Framing biosafety and biosecurity improvements as part of shared capacity-building and health-system initiatives can also make international engagement more relevant and attainable across diverse national contexts.

Oversight complications of new technologies

Rapidly advancing technologies are further complicating HCL management as they reshape laboratory practice itself. Automation and robotics reduce the need for direct human contact with pathogens, while AI systems are increasingly used to support routine monitoring, tracking of equipment performance and analysis of biological data. Synthetic biology and DNA-design tools allow laboratories to generate novel genetic material on-site, while advanced storage systems expand the range of biological agents that can be stored and reproduced.¹⁰ These tools reduce direct human involvement, thereby potentially reducing the ability to trace how biological materials are being used. Traditional assurance mechanisms such as facility registration or periodic self-reporting remain essential, yet they were designed for more static research environments and struggle to capture dynamic, real-time laboratory activity in today's increasingly distributed research systems.

In the past decade, discussions within the biological weapon prohibition regime have consistently reflected concern about the rapid pace and accessi-

⁹ Biological Weapons Convention, Working Group on the Strengthening of the Convention, 'Proposal for the establishment of an international platform dedicated to biosecurity and biosafety: SecBio', Working paper submitted by France, Senegal and Togo, BWC/WG/5/WP.12, 4 Dec. 2024; Biological Weapons Convention, Working Group on the Strengthening of the Convention, 'Article X report: Update on Australia's regional capacity building: Health security initiatives for the Indo-Pacific', Working paper submitted by Australia, BWC/WG/6/WP.4, 12 Aug. 2025; Biological Weapons Convention, Working Group on the Strengthening of the Convention, 'Proposal for the design of a plan to strengthen measures related to scientific and technological developments relevant to the Convention, with a focus on biosafety, bio custody and biosecurity', Working paper submitted by Chile, BWC/WG/6/WP.5, 18 Aug. 2025; and World Health Organization, World Health Assembly, 'Strengthening laboratory biological risk management', Resolution WHA77.7, 1 June 2024.

¹⁰ Brockmann, K., Bauer, S. and Boulanin, V., *Bio Plus X: Arms Control and the Convergence of Biology and Emerging Technologies* (SIPRI: Stockholm, Mar. 2019), pp. 4–25.



bility of emerging technologies.¹¹ States parties and experts assert that developments in AI, automation and digital biology are advancing faster than the mechanisms available to govern them.¹² However, this dialogue has predominantly focused on potential misuse and the erosion of barriers to weaponization, rather than on how these tools could be constructively applied to improve compliance or transparency. This research policy paper takes a complementary perspective—the following section examines how the same technological trends that heighten concern could also be harnessed to strengthen consistent, responsible laboratory oversight and thus reinforce implementation of the BWC.

II. Emerging technology use cases for high-containment laboratories

Use case 1. Using AI and DLT to address risks associated with gene synthesis

Advances in gene synthesis—the design and production of synthetic DNA sequences—are lowering technical and financial barriers to creating and modifying genetic material.¹³ Portable sequencing tools, benchtop gene printers and cloud-based ‘biofoundry’ services enable researchers to create, modify and assemble genetic material in routine laboratory environments. This has expanded access to techniques once limited to advanced research institutions. While this diffusion supports innovation, it also raises the risk that genetic material related to dangerous pathogens could be manipulated without appropriate oversight. However, access to and expertise in gene synthesis remain uneven, particularly in low-resource settings where technical and financial barriers persist.

Oversight of gene synthesis

Most oversight of gene synthesis currently takes place at the point of purchase through commercial screening of DNA orders. The International Gene Synthesis Consortium (IGSC)—a biotechnology industry-led group of gene-synthesis companies and biosafety-oriented organizations that represents a majority of the global DNA synthesis market—has developed best practice screening protocols.¹⁴ These protocols guide gene-synthesis companies in screening purchase requests against known pathogen samples and in reporting any anomalies that they detect. However, some providers still operate outside the IGSC framework and these protocols are not legally binding. Indeed, no country has implemented a comprehensive statutory DNA-screening requirement at the national level, and most existing guidance remains voluntary in practice, as with the United Kingdom’s ‘baseline

¹¹ Revill, J., Anand, A. and Persi Paoli, G., *Exploring Science and Technology Review Mechanisms Under the Biological Weapons Convention* (UN Institute for Disarmament Research: Geneva, 2021).

¹² Biological Weapons Convention, 9th Review Conference, ‘New scientific and technological developments relevant to the Convention’, Background information document submitted by the Implementation Support Unit, BWC/CONF.IX/7, 13 Dec. 2022.

¹³ Wadhwa, V., ‘The genetic engineering genie is out of the bottle’, *Foreign Policy*, 11 Sep. 2020.

¹⁴ International Gene Synthesis Consortium (IGSC), ‘Harmonized Screening Protocol v3.0’, 3 Sep. 2024.

expectations' issued in 2024.¹⁵ In 2025 the United States rescinded plans to implement a framework requiring genetic material screening for companies receiving government funding.¹⁶ Thus, in the absence of clear national or international requirements, screening practices remain uneven. While national regulations could help close these gaps, without coordination or shared standards differences across jurisdictions may still limit consistency.

Preventing irresponsible or malign gene-synthesis activities therefore relies primarily on voluntary compliance by researchers and institutions, guided by professional ethics and national biosafety frameworks. While some laboratories follow international standards such as ISO 35001—which sets out requirements for managing biological risks through structured policies, responsibilities and performance evaluation—others operate under informal or inconsistent biosafety guidelines, if any.¹⁷ Broader implementation of ISO 35001 has been constrained by its voluntary status, the costs of certification, and the absence of alignment between the standard and most national regulatory frameworks.¹⁸

The BWC's CBM E nominally covers national legislation, regulations and other measures related to biological agents and toxins. However, its focus is on pathogens and toxins, rather than synthetic DNA or gene-synthesis activities. It does not request governments to indicate whether HCLs are conducting gene-synthesis research or procuring these services through commercial providers. This reflects the historical framing of the CBM E questionnaire, which was designed before such technologies emerged. More broadly, the CMB survey system as a whole does not fully capture how states oversee or procure synthetic DNA, leaving uncertainty as to whether these activities—often involving non-pathogenic genetic fragments—fall within its intended scope. Without additional reporting or clarification, there is limited insight into national governance of gene synthesis and the extent to which HCLs are engaging in these practices.

AI and DLT to reinforce oversight

Several pilot efforts demonstrate how responsible research oversight can be reinforced by AI and by DLT—a class of digital systems, including blockchain, which store data across a decentralized network of computers, creating tamper-evident, timestamped records. AI can aid manufacturers in detecting irregular or suspicious DNA orders, while DLT-enabled platforms can store tamper-resistant transaction records showing when and how genetic material is ordered, shipped and used.

SecureDNA, a non-profit consortium based in Switzerland, has implemented an AI-enabled screening platform designed to identify potentially harmful DNA sequences.¹⁹ It can even identify those that have been subtly altered through modification or fragmentation to evade conventional screen-

¹⁵ British Department for Science, Innovation and Technology, 'UK screening guidance on synthetic nucleic acids for users and providers', 8 Oct. 2024.

¹⁶ The White House, *Winning the Race: America's AI Action Plan* (White House: Washington, DC, July 2025).

¹⁷ International Organization for Standardization (ISO), 'Biorisk management for laboratories and other related organisations', ISO 35001:2019, edn 1, Nov. 2019.

¹⁸ Millett (note 4); and Lentzos, F. and Koblenz, G. D., 'High consequence bio labs: Growing risks and lagging governance—Release of Global BioLabs 2023 report', 16 Mar. 2023.

¹⁹ SecureDNA, 'Frequently asked questions', [n.d.].



ing methods. The platform screens and flags suspicious DNA orders internally but does so without disclosing customer research details or company order data to third parties. It is accessible to verified gene-synthesis providers at no cost.

Complementing these efforts, Twist Bioscience, a US company working in partnership with Microsoft and IGSC members, announced in October 2025 that it is incorporating AI-enabled anomaly detection into its workflows for the screening of commercial transactions.²⁰ This change is based on pilot studies conducted since 2023.

While current DNA-screening systems primarily rely on AI-based anomaly detection and private databases, similar principles could be extended through DLT to provide verifiable audit trails for gene synthesis transactions. To have a meaningful impact, however, measures incorporating AI or DLT must be embedded within national oversight systems—for instance, through procurement, licensing or certification requirements—to ensure that synthesis providers and equipment manufacturers operate under consistent biosafety and biosecurity expectations. These private-sector initiatives highlight how technological tools can complement state responsibilities under the BWC by reducing misuse risks and strengthening accountability.

To extend such benefits more broadly, BWC states parties could use the Working Group on the Strengthening of the Convention as a forum to consider voluntary reporting templates or guidance that capture emerging areas such as oversight of gene synthesis. These formats would be completed by states parties but could incorporate information provided by laboratories or companies through national authorities. Establishing such channels would help ensure that transparency mechanisms keep pace with technological change. In turn, digital oversight tools could support more standardized, interoperable national reporting, which would reinforce implementation of BWC articles III (on transfer controls) and IV (on national implementation) and promote greater confidence in compliance.

Use case 2. Using AI and DLT to strengthen digital data integrity and laboratory information-management systems

Digital record-keeping has become central to high-containment research. It underpins experiment tracking, material inventories and biosafety documentation. Laboratory information-management systems (LIMS) and electronic laboratory notebooks improve efficiency by standardizing documentation and reducing manual error.

Information management in high-containment laboratories

When applied within HCLs, LIMS introduce security challenges related to the loss, theft or manipulation of stored data. If data can be modified without detection or if access is not properly controlled, it becomes difficult to verify whether biosafety procedures were implemented correctly or whether biological materials were responsibly managed.

Many high-containment facilities still rely on internal databases and tracking or incorporate mixed paper and digital records. This complicates

²⁰ Wittmann, B. J. et al., 'Strengthening nucleic acid biosecurity screening against generative protein design tools', *Science*, 2 Oct. 2025.

independent verification, the tracking of responsibility or detecting tampering.

Digital records are typically held within institutions and are rarely available for external review, which limits their value for accountability or confidence-building.²¹ Moreover, there is no global standard for data retention, validation or integrity checks across high-containment environments.

AI and DLT to manage information

Emerging technological advancements offer ways to bridge the accountability gap. Several commercial platforms have developed AI-enabled LIMS to standardize laboratory documentation. While many programs increasingly advertise being ‘AI-ready’, new systems have embedded AI directly into the system architecture to enhance traditional LIMS functions or provide new, intelligent capabilities. AI can be used to analyse metadata from LIMS platforms to detect irregular data-access patterns or discrepancies in sample-handling logs, to flag potential insider misuse or to identify record inconsistencies.²²

DLT can be implemented to secure information in HCLs by incorporating user authentication, access controls and automated audit logs. This makes it harder to alter records without leaving a trace. These laboratory activity audit logs are inherently tamper-evident, as any attempt to alter or destroy a data set is recorded and linked to a specific user and time.²³

DLT for transparency in research

Large research institutes often manage extensive collections of pathogen samples that circulate between laboratories, sample repositories and storage facilities. DLT can help to link physical transfers with digital records by creating verifiable chains of custody, thereby strengthening biosafety controls and auditability.²⁴ This technology can also link experiment results with specific protocols, storage conditions and user actions. While these systems are intended for operational efficiency rather than data security, they can reduce opportunities for data manipulation and create more transparent research environments.²⁵

However, these tools still function largely at the level of the individual facility, and their data outputs are not easily comparable across laboratories or national systems. This limits their use for broader transparency or assurance purposes. In addition, balancing data traceability with protection of intellectual property and commercially sensitive information remains a key challenge in designing DLT systems for laboratory use.

Nevertheless, these developments indicate that digital data systems are evolving from management tools to mechanisms for reinforcing responsible

²¹ Muluneh et al. (note 4); and Lentzos and Koblenz (note 18).

²² Benchling, ‘Benchling reveals the next chapter of R&D at Benchtalk 2025’, Press release, 14 Oct. 2025; Scispot, ‘alt-LIMS: Alternative for lab information management systems’, [n.d.]; and Kitishian, D., ‘Thermo Fisher Scientific’s AI strategy: Analysis of dominance in life sciences’, Klover.ai, 20 July 2025.

²³ ServBlock, ‘How it works: Secure, spoof-proof, and tamper-proof location data’, [n.d.].

²⁴ Ortiz-Lizcano, M. I. et al., ‘Increasing the security and traceability of biological samples in biobanks by blockchain technology’, *Computer Methods and Programs in Biomedicine*, vol. 231 (Apr. 2023).

²⁵ Benchling (note 22).



research oversight. Together with common reporting formats, these tools can support more attributable documentation of laboratory operations under the BWC's CBMs.

Use case 3. Using DLT to enhance standards for pathogen repositories and material traceability

HCLs often maintain or use pathogen repositories that support development of and training and research on diagnostics and medical countermeasures. These well-characterized biological materials serve as reference strains, allowing laboratories around the world to calibrate testing and measurement methods and to validate the results against common standards. During outbreaks, repositories support rapid response by supplying verified samples for vaccine development and therapeutic testing. Centralized collections also reduce the need for laboratories to acquire materials independently from nature, which reduces biosafety risks and research duplication. These collections thus constitute essential scientific infrastructure.

Risk of pathogen misuse

There is no global standard for documenting the origin, movement or use of these pathogen samples; unlike, for instance, the movement of chemical precursors controlled by the 1992 Chemical Weapons Convention (CWC).²⁶ As a result, materials may be shared or transferred without full authorization or traceability, and in some cases they can be duplicated without clear accountability, creating uncertainty about their intended final use. Some of these risks can be mitigated through structured access systems such as the European Virus Archive Global (EVAg), which provides controlled distribution of high-risk pathogens to verified research institutions under standardized biosafety and biosecurity conditions.²⁷ In the United States, the BEI Resources repository distributes regulated biological materials to approved institutions.²⁸ Both repositories generally offer access while only charging users for shipping costs.

Risk of misuse is compounded by ongoing disputes over international access and benefit-sharing (ABS) guidelines. This includes states' differing interpretations of the 2010 Nagoya Protocol—a supplement to the 1992 Convention on Biological Diversity that was designed to ensure that a country receives fair benefits when its biological resources are used for research or commercial purposes.²⁹ The protocol's application to pathogens in public health emergencies remains contested by some states parties, impeding cooperation and limiting transparency.³⁰

²⁶ Convention on the Prohibition of the Development, Production, Stockpiling and Use of Chemical Weapons and on their Destruction (Chemical Weapons Convention, CWC), adopted 3 Sep. 1992, opened for signature 13 Jan. 1993, entered into force 29 Apr. 1997, Verification Annex, parts VI–IX.

²⁷ European Virus Archive Global (EVAg), 'The EVA project: Concept and approach', [n.d.].

²⁸ BEI Resources, 'About: BEI Resources', [n.d.].

²⁹ Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization to the Convention on Biological Diversity, adopted 29 Oct. 2010, opened for signature 2 Feb. 2011, entered into force 12 Oct. 2014.

³⁰ World Health Organization, Executive Board, 'The public health implications of implementation of the Nagoya Protocol', Report by the director general, EB/148/21, 6 Jan. 2021.

AI and DLT to manage materials

Emerging technologies could support better management of biological materials without requiring direct monitoring. Chain-of-custody systems using DLT can apply unique digital identifiers to track pathogens as they move between laboratories. DLT can also provide tamper-evident records of transfers, storage and sample destruction. Meanwhile, AI can be used to assist in analysing and reconciling inventory data to pinpoint anomalies possibly linked to unauthorized access or movement.

Some pilot initiatives have demonstrated these possibilities and a proof-of-concept study was documented in 2023 using IBM's blockchain platform.³¹ The International Air Transport Association (IATA), an airline trade association, has piloted blockchain-based tracking for shipping pharmaceutical and temperature-sensitive goods, which may offer transferable lessons for the global traceability of pathogen material.³² In the European Union (EU), the TRACE project has piloted digital provenance tools for tracking the origin and transfers of samples, with results expected at the project's conclusion in 2026.³³

Extending such mechanisms could help strengthen implementation of articles III and IV of the BWC by improving accountability in how high-risk biological materials are managed and exchanged.

III. The way forward

The emerging technologies explored in the three use cases have largely been developed and deployed at the facility level; however, governments can incentivize or standardize their use through national biosafety frameworks and reporting formats. Consistent adoption of such practices can indirectly enhance states' internal capacity to monitor laboratory activities reliably and document biosafety practices consistently.

By strengthening national oversight infrastructure, technologies such as DLT traceability platforms, standardized reporting systems and AI-enabled documentation can help states parties meet their BWC obligations more effectively while building confidence in peaceful research intent. When integrated into transparency mechanisms such as CBMs, such systems could improve information quality and support technical cooperation among states parties.

Addressing disparities in capacity, cost and access will require not only institutional innovation but also practical incentives for participation. For low- and middle-income countries, strengthening transparency and responsible stewardship can demonstrate commitment to BWC Article X (on international cooperation). It can also contribute to a more cooperative environment in which confidence-building, technical exchange and capacity development are shared priorities among all states parties. Realizing this potential will require structured dialogue through existing BWC forums—particularly the Working Group on the Strengthening of the Convention—

³¹ Ortiz-Lizcano et al. (note 24).

³² International Air Transport Association (IATA), *Dangerous Goods Regulations*, 66th edn (IATA: Montreal, 2025), Section 3.6.2.

³³ European Commission, 'InTEgration and haRmonizAtion of logistiCs opErations', Horizon Europe Grant no. 101104278, CORDIS, 21 Apr. 2023.



where states, laboratory practitioners and technology developers can assess implementation experiences and identify where voluntary convergence around common practices is feasible.

Voluntary alignment among states parties, research institutions and industry partners around shared expectations for digital documentation, provenance tracking and responsible research disclosure could strengthen collective compliance with the BWC and reduce uncertainty between states. Although existing BWC forums provide a foundation for dialogue, meaningful progress on digital documentation and provenance tracking is likely to require a dedicated scientific and technological mechanism, similar to the Scientific Advisory Board of the Organisation for the Prohibition of Chemical Weapons (OPCW), or a coordinated technical platform. The precise form of this arrangement—whether an expert group or a standing forum—remains under discussion in the Working Group on the Strengthening of the Convention. However, its remit would be to consolidate technical advice, identify best practices and facilitate pilot projects on emerging technologies relevant to biosafety, biosecurity and verification support. It would then provide sustained, evidence-based input to states parties as they consider options for strengthening the BWC. Sustained engagement between scientific and policy communities will be essential to ensure that advances in bioscience reinforce, rather than undermine, the international prohibition on biological weapons.

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SIPRI RESEARCH POLICY PAPER

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